

Deep Learning Techniques for Brain Tumor Detection

Deep learning uses neural networks with multiple processing layers for medical image analysis.

Convolutional neural networks (CNNs) can learn spatial features from brain MRI images and assist with cla

Transfer learning adapts pretrained models to medical imaging tasks and can be useful when labeled datase

Image segmentation assigns labels to image pixels or regions. U-Net and its variants are commonly used fo

Performance can be evaluated using accuracy, sensitivity, specificity, precision, recall, F1 score, and R

Limitations include dataset size, class imbalance, image quality, scanner differences, labeling quality,